

ORIGINAL ARTICLE

Communication barriers to optimal access to emergency rooms according to deaf and hard-of-hearing patients and health care workers: A mixed-methods study

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Abstract

Background: This study aimed to identify communication barriers between health care workers (HCWs) and deaf and hard-of-hearing (DHH) patients. Both perspectives are offered to provide a comprehensive understanding.

Methods: Two consecutive studies were conducted from 2018 to 2021. Study 1 comprised mixed methods, employing a cross-sectional survey ($n=288$) and in-depth interviews ($n=9$) with DHH participants, utilizing accessible tools including sign language. Study 2 involved a cross-sectional survey of health care emergency workers without hearing loss ($N=391$).

Results: The perceived self-efficacy of DHH patients, and not their hearing loss, was linked with their ability to communicate independently with HCWs. No significant differences in successful communication with these providers were found vis-à-vis mode of communication utilized (sign language, writing, interpreter, etc.). In the qualitative findings, DHH patients noted two urgent care barriers: HCWs' communication unfamiliarity and patients' communication accessibility issues. Quantitative findings indicated a main barrier: difficulties in communicating with HCWs in general (57%) and specifically in the emergency room (ER; 65%). Only 28.8% reported being able to independently communicate with ER staff. Health care providers were not familiar with effective communication strategies when treating these patients. Respondents indicating that communication was not a barrier to care were mainly administrative staff (54.55%), compared to nurses (32.74%) and physicians (22.58%).

Conclusions: Communication solutions are needed to improve access to health services, especially in emergencies. Providing medical staff training on effective communication strategies with these patients could simplify interactions and

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reduce the reliance on hearing family members, potentially improving medical care. Implementing a communication policy for frontline staff, along with the use of visual aids, is crucial. Health care professionals may not realize that small changes can greatly improve communication with DHH patients.

KEYWORDS

accessibility, communication, deaf and hard-of-hearing patients, emergency room, health care workers

INTRODUCTION

Despite accessibility rights, laws, and regulations, deaf and hard-of-hearing (DHH) patients still experience difficulties. The main challenge involves communication with the hearing public, which may escalate during emergency circumstances.¹ This exposes DHH patients to potentially dangerous scenarios.²

In this paper, the term DHH patients refers to individuals with varying levels of hearing loss. DHH patients constitute a heterogeneous group with diverse needs and expectations. Perspectives can vary among oral deaf individuals, hard-of-hearing people, late-deafened persons, and sign language users, encompassing differences in communication preferences, cultural identification, and attitudes toward technological interventions.^{3,4} Sign language users typically regard deafness as a cultural and linguistic identity, prioritizing visual communication and Deaf culture.^{5,6}

The World Health Organization estimates that roughly 466 million individuals have hearing loss.⁷ Hearing loss can impede the ability to receive and transmit verbal information. Communication needs vary depending on hearing loss level, cultural health beliefs, preferred languages, health literacy, and other factors.^{8,9} Poor communication between DHH patients and health care workers (HCWs) impedes treatment, and risks arise from patients' limited understanding of their medical conditions and treatments.¹⁰⁻¹² Moreover, some personal characteristics, such as patient stress and anxiety, often heightened in medical contexts, can further impede comprehension, exacerbating the risk of miscommunication and compromising the quality of patient care.⁹ A low sense of self-efficacy can also act as a communication barrier. However, the use of sign language-adapted explanations offered to DHH patients showed improvement in self-efficacy reducing miscommunication.¹³ The presence of families may affect patient privacy, altering information exchange dynamics, and emotional involvement may lead to distorted details. Additionally, family members might lack sufficient proficiency in sign language or medical terminology, impacting effective communication.^{3,14}

Alternative means of communication—including sign language interpreters, lip reading, writing, apps, and/or graphic materials—exist, but each has its limitations.^{15,16} For example, interpreter costs and limited availability in emergencies result in overreliance on hearing family members.^{1,17,18} Lip reading, while a valuable communication tool for some DHH individuals, poses significant limitations in accuracy, particularly when facing unfamiliar or specialized

terminology such as medical vocabulary, potentially leading to critical misunderstandings in health care.^{19,20} Surgical masks impede the ability to read lips, and their mandatory use during the COVID-19 pandemic^{21,22} further intensified communication barriers between patients and HCWs.²³ Video remote interpreting (VRI) services is a relatively common practice when a professional interpreter is not present. Despite their increasing availability via smartphones,⁴ VRI services pose challenges, including technological internet issues and communication management between participants, that impact the overall efficacy of video-based interpretation.²⁴

The term “accessibility” indicates a process by which all services provided to the public are adapted for people with disabilities.^{9,25} Accessibility should not be taken for granted even within publicly funded health care systems. Israel has such a system with accessibility regulations grounded in the Equal Rights for Persons with Disabilities Law,²⁶ currently in the final draft stage.²⁷ DHH individuals are entitled to financial assistance for professional translation/transcription services, covering diverse areas, including 70h/year of in-person or online sign language translation. Although the registration rate for online translation services is high for self-identified Deaf individuals (42.86%), it is only 14.29% for hard-of-hearing people. The overall utilization of these services remains relatively low, particularly in health care (27%).²⁸ However, DHH patients encounter challenges accessing urgent health services despite existing laws.^{9,25}

The presence of a professional interpreter or a hearing acquaintance also increases access to medical information and improves quality of care.^{18,29} Nevertheless, emergency medical services (EMS) personnel have difficulty understanding and providing information to DHH patients^{1,30} who are more likely to use the emergency room (ER) than non-DHH patients.³¹ Furthermore, emergency HCWs often face challenges in communicating with Deaf American Sign Language (ASL) users, leading to patient frustration² and resulting in lower hospital admission rates and reduced implementation of routine ER care practices.³²

DHH individuals encounter numerous adverse societal circumstances that elevate their susceptibility to unfavorable health outcomes and increase their utilization of EMS.³³ The lack of research on ER use by DHH populations hinders the development of interventions for improving patient health and outcomes.³⁴ A comparative study examining ER utilization patterns among DHH patients and non-DHH English speakers yielded several significant findings. Notably, a marked disparity emerged in the length-of-stay metric.

DHH patients using ASL had a 9% longer ER stay than non-DHH, English-speaking patients, a mean increase of 30 min.³⁵ These results underscore potential communication barriers or systemic inefficiencies disproportionately affecting DHH patients in emergency care. This article explored communication barriers between HCWs and DHH patients, synthesizing insights from both groups to enhance understanding and improve emergency care.

METHODS

Study setting and population

Utilizing a mixed-methods approach our research,³⁶ spanning 2017 to 2021 in Israel, seamlessly integrated two studies. In Study 1, the initial study, we employed a sequential explanatory mixed-methods design.³⁷ First, we quantitatively gauged DHH patients' self-efficacy in communicating with HCWs; then, in the ensuing qualitative phase (of Study 1), through in-depth interviews, we gained further insights, uncovering needs, challenges, and barriers in health care situations. The integration of Study 1 phases is shown under Results, showcasing the synergy of quantitative and qualitative perspectives. This approach, chosen due to the underexplored nature of the field, facilitated the identification, examination, and population comparisons of phenomena, opening avenues for new research directions.³⁸ Study 1 was part of a comprehensive doctoral thesis research endeavor.

Study 2, a cross-sectional examination, involved HCWs (physicians, nurses, and paramedical and administrative staff) without hearing loss. We investigated barriers, available resources, and health care providers' perceived capabilities in addressing DHH patients' emotional, physical, and technical needs. The results of Studies 1 and 2 were fully integrated, reported in the last section under Results.

Ethical considerations

This study received approval from the Ben-Gurion University of the Negev's university review board (ref. no. 2016-11, 17 February 2016).

Data collection, participants, and recruitment

Study 1: DHH participants

Quantitative approach

A cross-sectional survey of 288 DHH participants (including those with hearing aids and/or cochlear implants) aged 18–60 years was conducted in various regions of the country. The survey instrument was validated for face and content validity in a previous study,³⁹ and content validity was reaffirmed by a five-expert panel to ensure

accessibility of the Qualtrics sign language videos. This sample represents approximately 0.52% of the estimated 55,700 DHH individuals aged 18 years and above in Israel.⁴⁰

Participants were recruited through DHH organizations, social clubs, and websites. The main researcher, a child of deaf adults, proficient in sign language, explained the study's goals and purposes, using sign language. She visited clubs in person and guided survey access and smart device use (smartphones and tablets), improving response rate and participant diversity. The snowball sampling⁴¹ was also employed, allowing participants to invite their DHH acquaintances to join the study. The survey utilized accessible language, simple phrasing, and sign language videos for self-administration via Qualtrics. It assessed participants' HCWs interactions, self-efficacy, communication methods, and urgent care access. The survey comprised 45 questions divided into four sections: demographics, health care system-related inquiries, a self-efficacy assessment, and coping in emergencies.

Qualitative approach

Based on the results of the above-mentioned survey and a thorough literature search (including Google Scholar, PubMed, literature regarding law and regulation, social media, etc.), semistructured interviews for DHH participants were developed, aiming to explore in greater depth some of the issues that emerged from the quantitative part of Study 1 and the literature survey.⁴² Specifically, after the quantitative phase, we employed snowball sampling⁴³ and conducted in-depth interviews with nine DHH participants. Privacy and anonymity concerns led us to abandon planned video interviews in favor of note-taking during sign language conversation. This allowed the main researcher to maintain a flow while capturing essential information without interruption. After each session, the researcher promptly documented recalled information, providing an organized, reflexive analytical tool. Participants were assigned pseudonyms to ensure anonymity. Participants, aged 23–58 years, represented diverse backgrounds and regions (Jews, Muslims, Christians, immigrants, and native-born from different parts of the country). Five of the nine were women. Participants were offered interviews in spoken language, Hebrew, with or without sign language, based on their preference, and interviews commenced upon obtaining their written informed consent and verbal or sign language permission. Interviews average 75 min. Data collection ceased upon reaching data saturation between December 2018 and May 2019.

Study 2: HCWs—quantitative approach

Study 2 conducted a cross-sectional assessment of HCWs in nonhospital and hospital settings from 2018 to 2021. The survey instrument was developed for Israeli HCWs in 2014 and had prior successful use across diverse HCW populations in Israel.^{30,44}

The survey focused on interactions between emergency HCWs/hospital staff and patients with varying ranges of hearing loss, and

comprised 20 items covering HCWs' demographics (age, gender, profession, work experience), professional interactions with DHH patients, and perceived capabilities. The questions included communication methods, training received, services offered, understanding of sign language, willingness to learn, and perceived barriers to providing adequate care. Ratings were based on a 3-point scale—1 (yes a lot), 2 (yes a little), 3 (not at all)—and a 5-point Likert scale for perceived capabilities, ranging from 1 (I have no ability) to 5 (I have a great deal of ability). A survey was administered to 1026 participants in urgent care access training for assisting DHH patients. The HCWs were employed at 11 general hospitals, two primary care clinics, the national training center for EMS workers, and two nursing schools. The sample encompassed physicians, nurses, paramedics, administrative staff, and ambulance drivers without hearing loss. The study design is presented in Figure 1.

Data analysis

Study 1

DHH participants—quantitative approach

Data were collected using Qualtrics software and exported to SPSS files (ver.27.0). Quality assurance measures and tests for normality of distribution were conducted prior to analysis. Descriptive data included disability type, communication methods, participation in DHH clubs, and demographic details (age, gender, marital status, education, religion). Descriptive statistics included means, standard deviations (SD), and percentages; chi-square tests of independence were used to investigate the relationship between nominal variables; and t-tests were used for comparing means between two groups of continuous variables. To examine the



FIGURE 1 Flowchart of the study's design. The first step was the quantitative survey for DHH patients and HCWs. The second step involved qualitative interviews with DHH patients. DHH, deaf and hard-of-hearing; HCWs, health care workers.

factors that were associated with the dependent variable, multivariable linear regression models were utilized. Specifically, the models incorporated three key predictor variables: participants' degree of independent communication with HCWs, their ability to handle emergencies, and their perceived self-efficacy related to managing these situations. The authors verified the requisite linear regression assumptions, including normality of residuals, homoscedasticity, linearity, and absence of influential outliers, through diagnostic checks to ensure that the modeling approach was appropriate.

Predictor selection was guided by theory, prior research, and pilot insights.⁴² Demographics (age, education, marital status) had established associations. Communication variables stemmed from past research and pilot data hypotheses.^{9,14} Self-efficacy perception was included as a predictor given its theorized influence on independence, corroborated by pilot findings. This multisource approach ensured that the regressions tested well justified a priori hypotheses.

In the first multivariable linear regression analysis, factors found to be significantly associated with the perception of self-efficacy were age, level of education, marital status, independent communication with hearing people, and independent communication with service providers ($F(6, 226) = 5.148, p < 0.001$). This model explained 10% of the variance in self-efficacy perception, with communication with service providers emerging as the strongest positive correlate. A second multivariable linear regression was then conducted, this time using independent communication with service providers as the dependent variable. The independent variables in this model included demographic factors, independent communication with hearing people, and the perception of self-efficacy ($F(6, 226) = 5.148, p < 0.001$). This model explained 25% of the variance in communication independence suggesting these factors had a meaningful impact on participants' levels of independent communication with service providers. Both regression analyses employed a direct (simultaneous) entry method, based on prior hypotheses about potential predictors.

DHH participants—qualitative approach

Data gathered from each interview were analyzed, and insights informed subsequent interviews. Purposive sampling was used when arranging participant descriptions, to ensure a broad range of supporting data. Dependability was assured through a code–recode procedure,⁴⁵ alongside a process of reflection and iteration until data saturation⁴⁶ was achieved. The main researcher transcribed all interviews, relying on her knowledge of sign language syntax and her familiarity with the “Deaf voice.” Sign language was predominantly used during interviews, with variations based on the interviewee's preference, ranging from a combination of signing and speaking to using only voice or articulating words without voice. Coding and analysis were conducted simultaneously with ongoing data collection. Researchers participated in peer-debriefing sessions to collaboratively analyze data—a process that involved systematic data coding. Interviews were conducted mostly in participants' homes

and DHH clubs, ensuring anonymity through name replacement and omission of unique identifying details. Reflexive thematic analysis was employed,⁴⁷ and Narratizer software facilitated transcript coding.

Study 2

HCWs—quantitative approach

We analyzed the results using SPSS (ver.27.0). Data analysis included chi-square tests and regression models. Variables incorporated into the regression models included HCWs' background characteristics such as age, gender, family status, profession, work experience, having a co-worker familiar with sign language, having a co-worker responsible for communication with DHH patients, availability of services at the organizational level, and prior training. Additionally, we assessed HCWs' communications with DHH patients and their perceived capabilities in treating this population. To explore the interrelationships among variables and predict HCWs' perceived ability to address the emotional and physical/technical needs of DHH patients, we employed multivariable linear regression. Covariates significant at $p < 0.01$ were included in the final regression models. This multivariable linear regression analysis unveiled how the independent variables, including demographic factors, previous experience with patients, work experience, and training, were related to and explained the variance in the dependent variable of addressing patients' emotional and physical/technical needs.

Furthermore, we utilized multivariable linear regression to assess HCWs' perceptions of barriers to providing adequate care due to communication difficulties. This regression modeling approach allowed us to determine the extent to which the independent variables accounted for differences in the dependent variable of perceived barriers from communication challenges.

RESULTS

Participants' characteristics

Study 1

DHH participants—quantitative approach

Study 1 comprised a survey with 288 DHH participants, including individuals using hearing aids and/or cochlear implants. Nearly two-thirds of DHH participants identified themselves as Deaf ($n = 183, 63\%$) and one-third as hard-of-hearing ($n = 91, 32\%$). Ten respondents reported that they had no limitation or disability (3.5%), although all the DHH participants met the criteria for being in the study. Most DHH participants were women ($n = 164, 66\%$), aged 38.93 years old on average ($SD \pm 12.57$ years, range 18–60 years). Most were native-born ($n = 183$), 74% were Jews ($n = 213$), 9% were Muslims ($n = 26$), and 17% were others (Christians, Druze, and atheists). Approximately 38.5% had a

university degree ($n = 111$), 17% had technical training ($n = 50$), and 28% had a high school diploma or basic education ($n = 75$). About 80% of the respondents indicated that they had a smartphone

($n = 231$). A summary of these data appears in Table 1. No significant relationship was found between the level of hearing loss (Deaf or hard-of-hearing), the use of sign language, and being

TABLE 1 Background characteristics of the DHH patients ($n = 288$) and HCWs ($n = 391$) survey from the two studies, including interviews of DHH patients ($n = 9$).

	Questionnaires		Interviews	Questionnaires	
	DHH patients ($n = 288$)		DHH patients ($n = 9$)	HCWs ($n = 391$)	
Age (years) mean-SD)	38.93 (± 12.57)		39.44 (± 0.66)	38.89 (± 13.39)	
Gender					
Female	164 (66)		5	260 (66.50)	
Male	86 (34)		4	122 (31.20)	
Hearing disability				Profession	
Deaf people	183 (63)	Without a hearing aid	3	Nurses and midwives	119 (30.59)
Hard of hearing	91 (32)	With a hearing aid	2	Paramedical and technical staff (laboratory, imaging, physiotherapists)	89 (22.88)
DHHP reported no limitations	10 (3.5)				
Cochlear implant					
With cochlear implant	44 (18)	With cochlear implant	1	EMS workers	88 (22.62)
Without cochlear implant	77 (71)	Without cochlear implant	6	Physicians	31 (7.97)
				Mental health workers	7 (1.80)
Will want one in the future	20 (7)	With hearing aid + cochlear implant	2	Work experience	
				10+ years	132 (33.5)
				6–10 years	48 (12.28)
				3–5 years	76 (19.44)
Had cochlear implant but do not use it now	8 (3)	Had cochlear implant but do not use it now	1		
Partner				Work unit	
With partner	183 (63.54)		7	Less than 2 years	124 (31.71)
Partner with hearing disabilities	148 (52.6)		6	Hospitals	244 (63.05)
Place of birth: Israel	183 (63.5)		7	EMS	75 (19.38)
Religion					
Jewish	213 (73)		7	Preventive and primary care services	38 (9.82)
Muslim	26 (9)		2	Others	30 (7.7)
Others	24 (7)				
No response received	(11)				
Educational status					
Academic education	111 (38.5)		4		
Postsecondary education	50 (17)		2		
Middle school and secondary school	75 (28)		3		
Smartphone users	231 (80)				

Note: Data are reported as mean ($\pm$ SD) or n (%).

Abbreviations: DHH, deaf and hard-of-hearing; HCWs, health care workers.

part of the Deaf community. The *t*-tests and corresponding *p*-values for all three variables exceeded 0.05.

DHH participants—qualitative approach

Nine participants with diverse levels of hearing loss were interviewed, aged between 23 and 58 years. They represented various backgrounds, including Jews, Muslims, immigrants, and native-born from different regions of Israel. Five of the DHH participants were women. Most had partners with hearing loss, whereas some had family members with hearing problems and others did not. Participants also varied in their level of social engagement within the Deaf community or DHH organizations.

Study 2

HCWs—quantitative approach

In our sample (*n*=136) almost 80% were women, aged 38.89 years old on average (*SD* ±13.39 years). More than one-third (131, 33.5%) had 10+ years of professional experience, and 124 (31.71%) had less than 2 years of experience. About one-third were nurses and midwives (*n*=119, 30.59%), 89 (22.88%) were paramedical and technical staff (e.g., laboratory, imaging, physiotherapists), 88 (22.62%) were EMS workers, 55 (14.14%) were administrative staff, 31 were physicians (7.97%), and seven (1.80%) were mental health workers. Most

participants (*n*=244, 63.05%) worked in hospitals, and about one-third (*n*=99, 33.56%) had no clinical contact with DHH patients.

Integrated findings from the studies DHH participants (Study 1) and HCWs (Study 2)

Integrated findings from the mixed studies with DHH participants and the HCWs survey five central topics:

Assessment of communication between DHH patients and HCWs

Of the HCWs who reported having contact with DHH patients, 38 (10.76%) reported that communicating with DHH patients was unsuccessful (no significant differences by gender, age, training, profession, or work experience). HCWs who were successful in communicating with DHH patients reported using the help of a hearing relative (38.38%), lip reading (50%), writing (26.77%), and sign language (8.86%). Only 4.8% used an interpreter. Most other interactions involved a combination of methods such as those mentioned above, speaking slowly or loudly, and/or hand and body gestures.

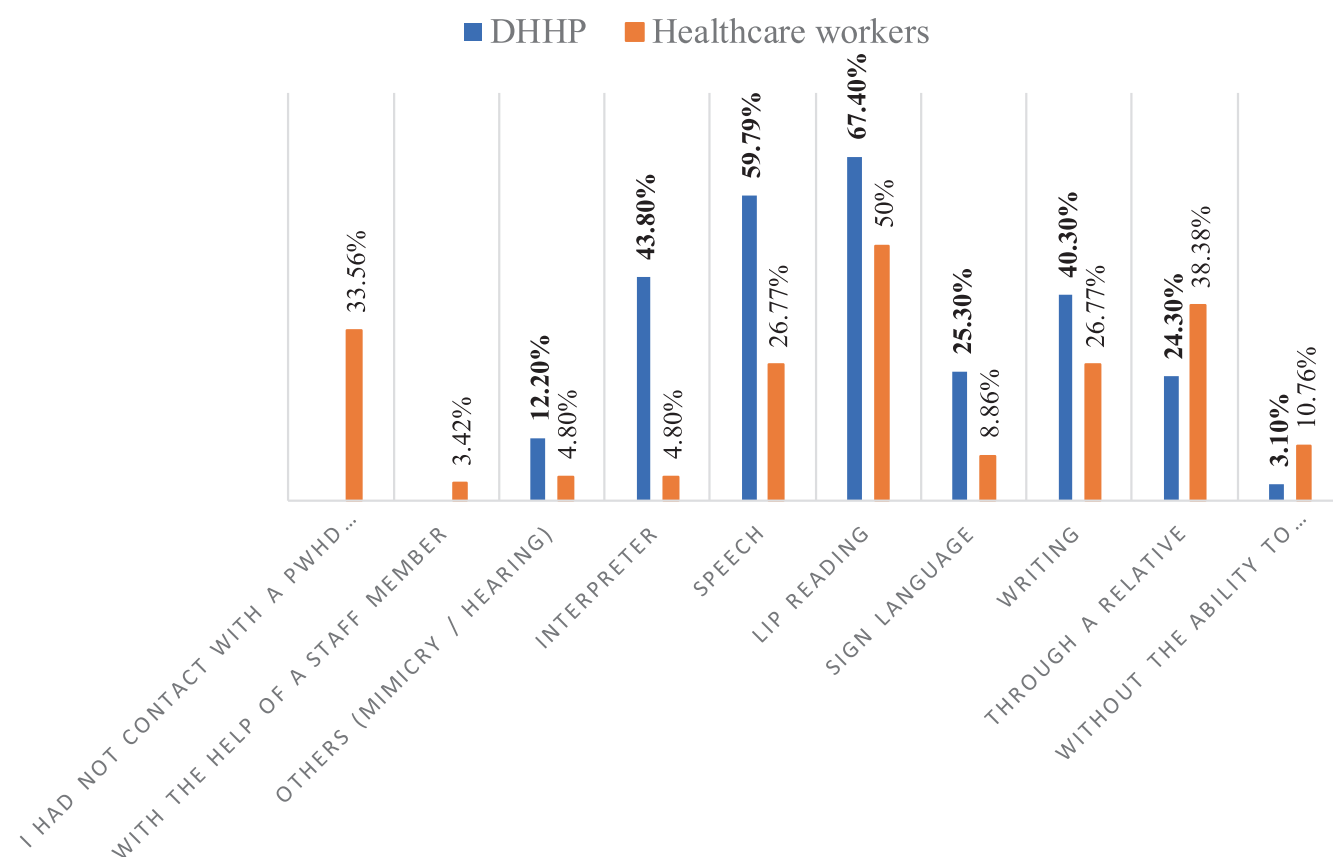


FIGURE 2 Comparison of forms of communication between DHH patients and HCWs. Lip reading was the most popular form of communication. DHH, deaf and hard-of-hearing; HCWs, health care workers.

Figure 2 illustrates DHH patients' survey responses. The most common communication method with hearing people was lip reading (67.4%). They also reported that a sign language interpreter was essential and helped them communicate (43.80%; Figure 2).

Communication barriers

Most HCWs ($n = 389$, 60.05%), regardless of gender, indicated that communication difficulties presented an obstacle in caring for DHH patients. However, these assessments varied according to occupation. Those reporting that communication difficulties were not an obstacle to care were administrative staff (54.55%) versus nurses (32.74%), EMS workers (26.14%), and physicians (22.58%; $p < 0.001$).

The prevalence of communication difficulties as an obstacle to caring for DHH patients was negatively associated with HCWs' work experience: 24% of providers with up to 5 years of experience reported communication difficulties, compared to 36.17% among those with more than 10 years of experience. Older respondents aged 50 years and above reported more communication difficulties (47.06% vs. 23.23% 50-, $p < 0.001$) than younger HCWs.

The qualitative funding indicated that DHH participants perceived communication barriers, reporting health care system, especially urgent care, were ill-equipped to meet their needs:

I'm married, right? Why do [the] physicians call my parents? It's an emergency, OK, why not send a message? Am I a little boy? The medical staff ... why do they call parents?!—

Muhamad with hearing aids. Sign language conversation with voice.

We needed to speak with the physician about our son [in the ER] ... but we did not understand; we could not read his lips, it was hard!! Nor did my wife understand him [also a Deaf person]. The physician spoke quickly, we did not understand, it was very difficult!!! Very difficult!!! —

Baruch without hearing aids. Sign language conversation. Sometimes with voice.

The quantitative data reinforce these findings, indicating that in the ER, only 28.8% of DHH patients felt that they could communicate independently with HCWs, whereas 33% felt that they could not communicate without assistance, and 31.6% did not know whether the urgent care personnel would understand them. Compared to physicians familiar to them (e.g., general practitioners), most (51%) could communicate independently, while only 17% reported needing assistance. The situation worsened when it came to mental health care (e.g., with psychiatrists, psychologists, and social workers), as 36.5% of the respondents reported that they could not communicate independently in this context (Figure 3).

Use of a professional interpreter

Among DHH participants, 15.3% reported always using interpreters when communicating with hearing people ($n = 41$), 11.2% used them once a week ($n = 30$), 40.7% used interpreters one or more times per month ($n = 109$), 22.8% did not use interpreters ($n = 61$), and 10% indicated that such use was not needed ($n = 27$). Of those with cochlear implants, most (57.6%) did not use sign language interpreters; 16.3% of the DHH participants with a cochlear implant exhausted their entire free quota of translation services; and 15.9% reported using an interpreter weekly, 15.9% monthly, and 13.6% quarterly.

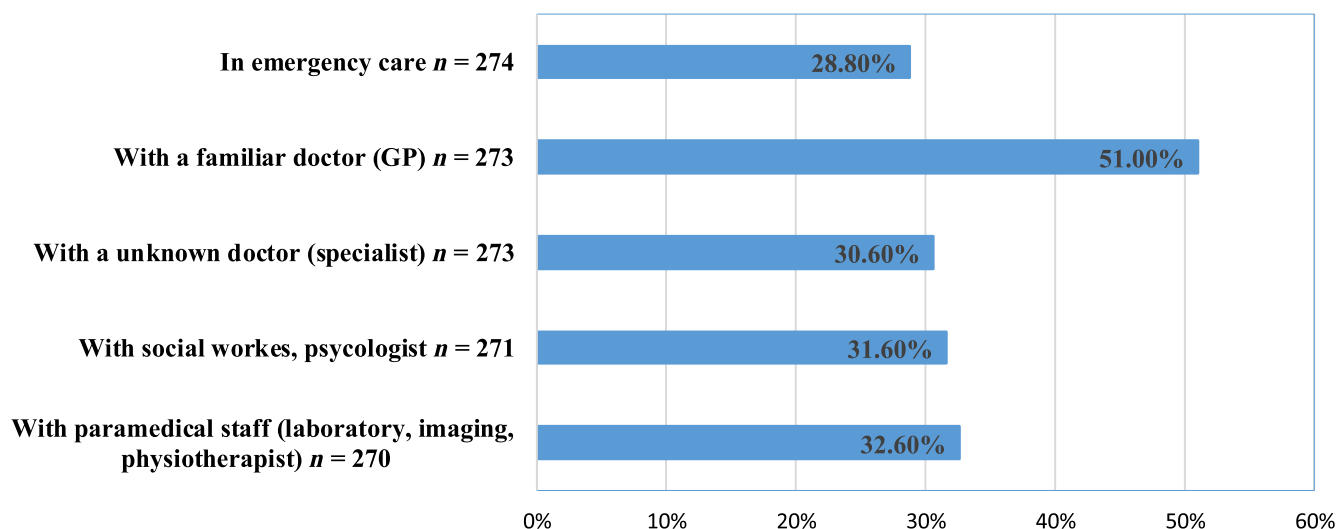


FIGURE 3 Communicating independently with different types of HCWs. Perception of independent communication among DHH patients is superior (51%) with a familiar doctor, such as a GP. DHH, deaf and hard-of-hearing; GP, general practitioner; HCWs, health care workers.

The DHH participants talked about problems in finding sign language interpreters in emergencies:

My son got sick and the physician sent us to the ER. I called [video-called] the interpreter first ... then my husband!!!! I sent a text for a taxi ... luckily it was possible through the Internet ... I had a hard time writing a message on my cell phone to find an interpreter while at the same time having my son in my arms! In the end, I found an interpreter, but the pressure of taking a child alone to the ER, trying to calm him down, and having to focus on finding an interpreter is inhumane!!! Why are there no interpreters in hospitals?!—

Sara without hearing aids. Sign language conversation without voice.

Participants with cochlear implants who used sign language also reported needing sign language interpreters:

We need more interpreters, a more accessible system; in emergencies, we should not have to look for an interpreter! —

Assaf with cochlear implants. Sign language conversation with voice.

The quantitative data from the HCWs reinforce these findings, in that only 14.63% of HCWs reported that their organization had translation services in sign language, and only 5.37% had a colleague who used sign language.

Participants' communication solutions

When I first arrived at the ER, I went to the secretary and I showed her that I'm a Deaf person, was written in the physician's letter. I wrote on my phone that I wanted to fully understand and asked the medical staff to respond in writing.—

Amal without hearing aids. Sign language conversation with Arabic words.

We need interpreters, but we need to teach Deaf people to read lips and speak because we live in a hearing world.—

Hanna with cochlear implant and hearing aid. Sign language conversation with voice.

The DHH participants described two barriers in urgent care: the fact that HCWs do not know how to communicate with DHH patients and the lack of accessibility to helpful communication services such as sign language interpreters in emergencies.

Perceived ability to assist emotionally or physically/technically

Only a small percentage of HCWs saw themselves as capable of responding to the emotional (9.83%) or physical/technical (28.17%) needs of DHH patients. An association was found between training in how to interact with DHH patients and HCWs' perceived ability to provide emotional care: with training, mean $\pm$ SD 2.07 ± 0.76 ; and without training, mean $\pm$ SD 2.05 ± 0.59 ($t(111.08) = 0.29$, $p < 0.01$). Using multivariable linear regressions, we found relationships between training, gender, place of employment (e.g., hospital, EMS), and occupation ($R^2 = 0.21$, $F(4, 367) = 4.25$, $p = 0.002$) and the ability to provide technical assistance. Associations were also found between the ability to provide emotional assistance and place of employment and age ($R^2 = 0.14$, $F(2, 353) = 3.43$, $p = 0.034$).

Furthermore, profession was significantly associated with perceived ability to deal with the emotional needs of DHH patients. Those who felt least capable in this area were EMS workers (4.6%) then physicians (10%), nurses (19.66%), and administrative staff (20%).

In the qualitative findings, DHH participants corroborated the results of the HCWs. The DHH participants reported feeling they did not receive emotional assistance from the physicians in the ER. The physicians did not seem to have the time or patience for such assistance in the ER, but other staff provided emotional support.

I was with my daughter, and the physician could not explain anything to me; he had no patience and asked a nurse for help ... she explained things to me in writing; she took the time and was patient. That's what we need ... patience.—

Rachel without hearing aids. Sign language conversation without voice.

Only 16.02% of the HCWs reported having training related to treating DHH patients. About one-third (31.46%) reported having organizational services for DHH patients, and nearly half indicated they did not know if their organization had special services for this population (47.37%). Almost half of the participants reported having accessible technical services such as websites and SMS capabilities (48.56%) in their organizations, and one-third indicated that their organization had an amplification system for DHH patients (31.28%).

DISCUSSION

Communication skills are vital for medical care, particularly in emergencies⁴⁸; they are also crucial for psychological or psychiatric treatment.³ As shown in the current study, ineffective communication among providers and health care systems during emergencies led to psychological harm for DHH patients. The loss of communication access systematically reduced these patients' capacity for collaborative decision making regarding their medical care, elevating the risk of health care delays.⁴⁹

Sign language interpreters (physical and online) can be helpful in cases where DHH patients can communicate in this way.^{28,50,51} Additionally, HCWs fluent in sign language improve the health care for these patients.⁵² The findings of Lyons and Normandin⁵³ suggest that ER nurses should adopt a proactive approach to educating themselves about sign language and ensuring access to resources for patients with DHH. Incorporating effective individualized communication strategies during emergency care can enhance patient outcomes and reduce miscommunication, errors, and mortality.

In this mixed-methods study, we identified several factors related to the accessibility to health care treatment for DHH patients, particularly in emergencies. Most HCWs (75%) had treated DHH patients, but over 80% reported no training about communication with this population. Adequate training of HCWs is essential to provide better care for these patients.^{54,55} The findings from a U.S. study involving Deaf adults using sign language revealed their ER encounters, compared to primary English users, were associated with a reduced likelihood of hospital admission and a lesser implementation of routine ER care practices.³² Health care systems are legally and ethically bound to uphold patients' communication access rights, as mandated by the Americans with Disabilities Act.⁵⁶ Israel's Equal Rights for Persons with Disabilities Law (1998, amended 2013), exemplifies this development. Its amendment mandates accessible health care services that preserve the dignity and privacy of individuals with disabilities.⁴⁸

From the perspective of DHH patients, the most important issue is the communication barrier with HCWs.^{10,57} HCWs also identified this issue as a major problem.^{30,58} They had difficulty transmitting information that might have been vital to these patients' treatment. Adequate information on all aspects is essential to enable patient participation in decision making,⁵⁹ and DHH persons often struggle with health care marginalization.⁵² In this study, less than 10% of the HCWs reported having a colleague who used sign language, a colleague responsible for interacting with DHH patients, or other services associated with DHH patients at the organizational level. Furthermore, more than half of the HCWs indicated that communication difficulties presented an obstacle to providing adequate care for DHH patients. Older health care respondents aged 50 years and above cited more communication difficulties (47.06% vs. 23.23% 50-, $p < 0.001$). This result may be explained by the increased use of communication technologies such as apps (Figure 4) among younger HCWs and the increased use of smartphones by DHH persons.⁴

Professional in-person interpreters are generally optimal for sign language users in health care settings. Given its cost-effectiveness and accessibility, VRI use should be expanded in the ER for DHH patients.⁶⁰ Though a viable alternative when in-person interpreters are unavailable, VRI faces challenges like connectivity issues and positioning difficulties in emergencies. It may also not accommodate preferences for interpreter gender or provide clear information about the quality of interpreter services. Nevertheless, VRI remains a valuable secondary option.

Our findings indicate that patients and HCWs agree communication is further hampered during emergencies. Nevertheless, in



FIGURE 4 The app facilitates basic communication with DHH patients in emergencies, supporting five languages (Arabic, English, Hebrew, Russian, and Israeli Sign Language). The creation of the app was financed by "Mashabim" The Community Stress Prevention Centre (CSPC), the Israel Trauma Coalition (ITC), The Institute for the Advancement of Deaf People in Israel, and the Ben-Gurion University of the Negev. DHH, deaf and hard-of-hearing.

routine situations, 51% of the DHH patients affirmed being able to communicate independently with a familiar physician.

One of the key factors that emerged from our findings was the need for training of health care providers in communicating with DHH patients. Training HCWs and teaching strategies for effective communication (Figure 5) with these patients can improve such patients' access to health services and treatment.^{56,58} Only one-quarter of the HCWs felt capable of meeting the emotional and physical/technical needs of DHH patients adequately or excellently. Thus, untrained workers saw themselves as less capable of assisting trained HCWs. Health care training programs should incorporate sessions and simulations designed to enhance communication skills with DHH patients.^{30,61}

Visual aids like charts, diagrams, brochures, and leaflets can enhance DHH patient education and communication in health care.⁶² This method effectively elucidates concepts, fundamental anatomical knowledge, procedures, and medication information. Supplementing visual aids with online resources and QR codes to facilitate the transmission of information would strengthen instructional efforts and comprehension.⁶³ Institutionalizing visual aids as policy is imperative, especially in the ER.

DHH patients predominantly utilized lip reading to communicate with hearing individuals. Although lip reading has a high error rate and is not the most effective method of communication, it remains the most used method due to limited accessibility to other

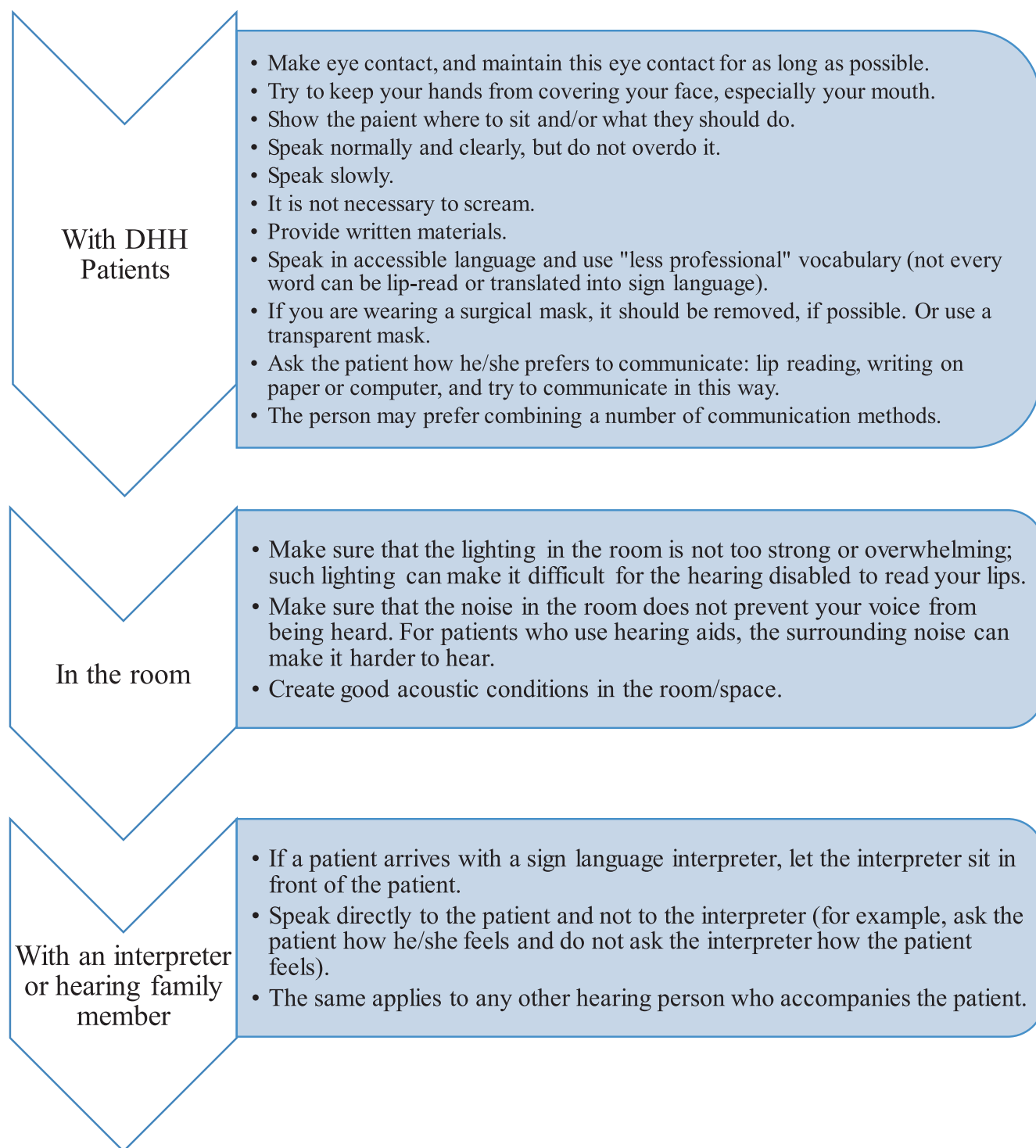


FIGURE 5 Guideline recommendations. Strategies for effective communication with DHH patients.

options. We recommend establishing an effective communication office policy wherein frontline staff inquire about the communication needs of DHH patients and document these requirements in patients' medical records for future reference. Additionally, creating a database of trained sign language interpreters with medical experience, whether in person or online, is advised to ensure accessibility in all future medical situations.

LIMITATIONS

This study has several limitations that warrant consideration. The 18- to 60-years age range may limit generalizability, warranting further research on older populations. Despite efforts to ensure accessibility, the digital survey may have been subject to potential misinterpretations of sign language due to intergenerational/

regional variations. The multistep translation process (Hebrew to Israeli Sign Language to English) posed challenges in finding precise equivalents. As with any translation, exact equivalents were not always feasible. Additionally, the HCWs sample drawn from communication workshop may exhibit selection bias. These limitations should inform result interpretation and future study design.

CONCLUSIONS

Communication barriers deprive deaf and hard-of-hearing patients of quality, even vital, medical care. In this research, we explored both sides of the interaction between deaf and hard-of-hearing patients and health care workers. Given the moral and legal obligation to care for everyone, we must understand the needs of this population and the medical staff to make recommendations and provide solutions that improve medical care.

Training HCWs on effective deaf and hard-of-hearing patient communication could mitigate complexities, reduce patients' reliance on hearing mediators, and improve care. Establishing frontline communication policies and visual aids is also necessary, as health care workers may not recognize minor modifications that can significantly enhance these interactions.

AUTHOR CONTRIBUTIONS

All authors contributed to the manuscript. All authors contributed to the design of the study. Dr. Tannenbaum-Baruchi conducted the data collection and analysis and drafted the manuscript. All authors contributed to the discussion section of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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